

Course 06: Data Analysis

Comprehensive Learning Guide & Practical Prompting Playbook

MODULE OVERVIEW

Harness AI for statistical interpretation, data cleaning strategies, SQL/Python query generation, data visualization planning, and turning quantitative metrics into narrative insights.

KEY LEARNING TOPICS

- Data Cleaning, Normalization, & Preprocessing Strategies
- SQL Query Construction & Optimization Prompts
- Python/Pandas Data Wrangling & Scripting
- Translating Metrics into Executive Dashboards & Stories

PROMPTING STRATEGY & ARCHITECTURE

Include schema definitions, sample data rows, and explicit mathematical definitions in the prompt to ensure calculations and queries are 100% accurate.

ROLE

You are a Senior Data Analyst and BI Specialist.

TASK

Write an optimized SQL query and a Python Pandas script to analyze customer churn from a SaaS sales database.

TABLE SCHEMA

Table: `user\_subscriptions`

Columns: `user\_id` (INT), `signup\_date` (DATE), `cancel\_date` (DATE), `monthly\_amount` (DECIMAL), `status` (VARCHAR)

REQUIREMENTS

1. Calculate monthly churn rate percentage for the year 2025.
2. Group results by month.
3. Write clean SQL (PostgreSQL dialect) with clear comments.
4. Provide equivalent Python Pandas code snippet for verification.
5. Highlight 3 potential business causes to investigate if churn spikes > 5%.

BEST PRACTICES & PRACTICAL TIPS

- Always feed explicit column names, data types, and sample data rows into your prompt.
- Ask the AI to explain its query logic step-by-step to catch edge cases (e.g., NULL values).
- Use the AI to write data validation test cases for your data pipelines.